

Efficient Heat Capacity of Methane-Loaded MOF-5 with Classical Molecular Dynamics and Machine-Learned Hessians

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September 12, 2026

Abstract

This report develops an efficient heat-capacity workflow for methane-loaded MOF-5 using a machine-learned interatomic potential (MLIP). Classical flexible-cell NPT molecular dynamics (MD) samples the loaded system and retains anharmonic methane and host-guest motion. Automatic differentiation (AD) through the matching PET-JAX model provides sparse Cartesian Hessians and normal-mode frequencies at optimized local minima. Rather than reproduce the reference simulations with an MLIP, the workflow applies the hybrid approximation proposed by Kapil *et al.*: the loaded-system classical anharmonic heat capacity is corrected by the difference between quantum and classical harmonic heat capacities evaluated for the same loaded composition and PET potential. The already-equilibrated empty MOF-5 structure is relaxed directly and evaluated with one reference Hessian; no empty-framework MD campaign is required. The resulting low-temperature and model-dependent accuracy must be validated for the selected MLIP. Finite-trajectory sampling error, minimum-to-minimum variation of the harmonic correction, and classical MLIP uncertainty from a calibrated last-layer prediction-rigidity (LLPR) ensemble are evaluated as distinct contributions.

1 Scientific motivation and scope

Metal-organic frameworks (MOFs) combine inorganic nodes with organic linkers to form porous crystalline networks. Their vibrational spectrum contains high-frequency local bond vibrations as well as soft linker, pore, and collective framework modes. Because the contribution of a quantum vibrational mode depends strongly on the ratio of its energy to $k_B T$, a broad frequency distribution produces a strongly temperature-dependent heat capacity. Guest adsorption adds translational, rotational, and host-guest degrees of freedom that may be weakly confined and strongly anharmonic.

The material considered here is MOF-5. Its conventional cubic cell contains 424 framework atoms with composition $C_{192}H_{96}O_{104}Zn_{32}$. In the loaded calculations, 100 methane molecules add 500 atoms, giving a 924-atom system with composition $C_{292}H_{496}O_{104}Zn_{32}$. The principal objectives are:

1. to sample the anharmonic thermodynamics of methane-loaded MOF-5 with classical NPT MD and independent replicas;
2. to compute force constants efficiently by sparse AD at optimized empty and loaded structures;
3. to obtain and converge the loaded-system harmonic quantum correction;
4. to combine that correction with the classical enthalpy derivative to estimate the loaded-system C_P efficiently; and
5. to establish a reproducible route from short cluster validation jobs to statistically meaningful production calculations.

The workflow has three computational stages. Loaded-system MD supplies the classical anharmonic enthalpy curve. Fixed-cell optimizations followed by AD Hessians supply quantum-minus-classical harmonic corrections. A final analysis combines the two terms with consistent composition and mass normalization. The empty-framework Hessian is retained as a reference and validation calculation, not as a replacement for the loaded-system correction.

The aim is to apply and assess this hybrid approximation on the PET potential, not to reproduce the numerical curves, force field, methane-insertion ensemble, or simulation protocol of Kapil *et al.* Agreement with a reported reference value can be a useful external comparison, but is not the criterion for success. The criteria are a well-defined, consistent PET potential across all terms, converged sampling and Hessians, and uncertainty estimates for the resulting hybrid curve.

2 Theory

2.1 Machine-learned interatomic potentials

2.1.1 Learning the potential-energy surface

Within the Born-Oppenheimer approximation, the electronic ground-state energy defines a potential-energy surface (PES) for the nuclei. An interatomic potential replaces an electronic-structure calculation by an approximation to this map,

$$(\mathbf{R}, \mathbf{h}, \mathbf{Z}) \mapsto U(\mathbf{R}, \mathbf{h}, \mathbf{Z}), \quad (1)$$

where $\mathbf{R} = (\mathbf{R}_1, \dots, \mathbf{R}_N)$ are the nuclear coordinates, $\mathbf{h}$ is the periodic cell, and $\mathbf{Z}$ lists the chemical species.

A machine-learned interatomic potential (MLIP) represents this map by a differentiable function with parameters θ , fitted to energies and forces from electronic-structure calculations. Modern local MLIPs usually write the extensive energy as

$$U_\theta(\mathbf{R}, \mathbf{h}, \mathbf{Z}) = U_{\text{base}}(\mathbf{Z}) + \sum_{i=1}^N \varepsilon_i(\mathcal{N}_i; \theta), \quad (2)$$

where U_{base} is an optional composition-dependent baseline and $\mathcal{N}_i$ is the learned environment of atom i . The decomposition makes the cost approximately linear in N at fixed density and makes the energy size extensive. The individual atomic energies are a model construction rather than uniquely defined physical observables.

The force field is obtained by differentiating the scalar energy,

$$\mathbf{F}_i = -\frac{\partial U_\theta}{\partial \mathbf{R}_i}, \quad (3)$$

so the predicted forces are conservative, provided the same differentiable energy model is used throughout. Cell derivatives similarly give the virial or stress when the model and its interface support them. Automatic differentiation can be applied a second time to obtain the Cartesian Hessian used later in this project.

2.1.2 From an atomic structure to a graph

A graph neural network (GNN) acts on an attributed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, following the notation of Bronstein *et al.* [1]. The node set $\mathcal{V}$ contains the $N = |\mathcal{V}|$ atoms. An input node signal is a map $\mathbf{x} : \mathcal{V} \rightarrow \mathbb{R}^s$; its value at node u is $\mathbf{x}_u$, and stacking all node features as rows gives $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_N)^T \in \mathbb{R}^{N \times s}$. The edge set encodes local interactions. Because distinct periodic images of one atom may enter the same environment, the atomistic graph is more precisely a directed periodic multigraph. An edge $(v, u, \mathbf{n}) \in \mathcal{E}$, directed from the image $(v, \mathbf{n})$ to the central atom u , is included when

$$\mathbf{r}_{uv\mathbf{n}} = \mathbf{R}_v + \mathbf{h}\mathbf{n} - \mathbf{R}_u, \quad (4)$$

$$r_{uv\mathbf{n}} = \|\mathbf{r}_{uv\mathbf{n}}\| < r_c, \quad (5)$$

where $\mathbf{n} \in \mathbb{Z}^3$ identifies a periodic image and r_c is a cutoff. The incoming one-hop neighborhood is therefore the indexed set

$$\mathcal{N}_u = \{(v, \mathbf{n}) : (v, u, \mathbf{n}) \in \mathcal{E}\}. \quad (6)$$

This is the periodic analogue of the neighborhood $\mathcal{N}_u$ in Ref. [1]; retaining $\mathbf{n}$ distinguishes repeated images of the same node. Relative displacements remove dependence on the absolute origin. A sufficiently smooth cutoff function $f_c(r)$, whose value and relevant derivatives vanish at r_c , suppresses a neighbor’s contribution as it leaves the environment. Smooth forces and Hessians require consistent cutoff treatment throughout the network, including attention normalization and energy readout, rather than continuity of the cutoff value alone. The representative single cutoff and the comparison of cutoff functions are shown in Figures 1 and 2, respectively; both are from the PET paper [5].

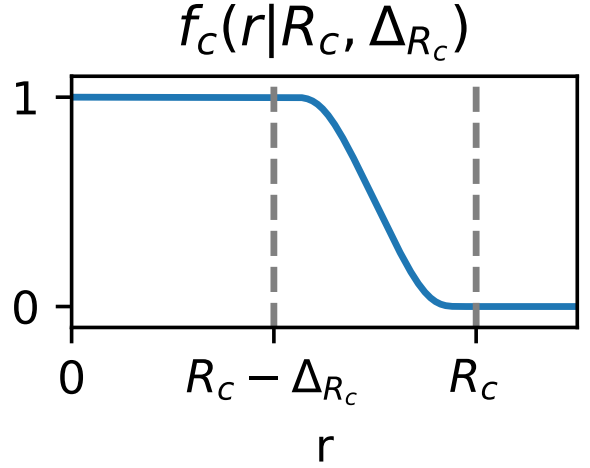


Figure 1: A smooth radial cutoff suppresses a neighbor contribution as its distance approaches the outer cutoff r_c . Smooth decay is needed for continuous energies, forces, and Hessians. Figure from the PET paper [5].

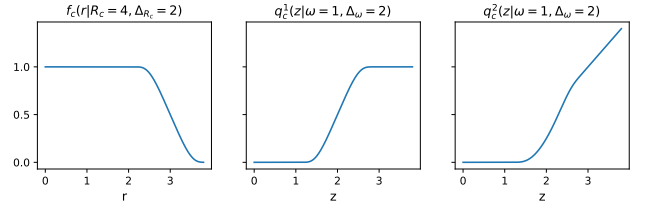


Figure 2: Comparison of radial cutoff functions used to attenuate neighbor contributions near r_c . The value and the derivatives relevant to the model must be treated consistently in every energy contribution. Figure from the PET paper [5].

Neural networks operate on fixed-width numerical representations. Chemical identity is therefore mapped from a discrete species label to a trainable embedding vector,

$$\mathbf{x}_u^{(0)} = \mathbf{E}_Z[\mathbf{Z}_u] \in \mathbb{R}^d, \quad (7)$$

where each row of the embedding matrix $\mathbf{E}_Z$ is learned during training. In the attributed-graph notation, geometry is carried by edge features. For example, a radial edge signal can be written

$$\mathbf{x}_{uv\mathbf{n}}^{\text{edge}} = f_c(r_{uv\mathbf{n}})[\varphi_1(r_{uv\mathbf{n}}), \dots, \varphi_K(r_{uv\mathbf{n}})], \quad (8)$$

and, in models that retain directional information, by the displacement $\mathbf{r}_{uv\mathbf{n}}$, its unit vector, spherical harmonics, or learned functions of these quantities. Here “embedding vector” means a vector in latent feature space; it need not transform as a physical three-dimensional vector.

Suppressing periodic-image labels for a moment, graph connectivity may also be represented by an adjacency matrix $\mathbf{A}$. A relabeling of the nodes by a permutation matrix $\mathbf{P}$ sends $(\mathbf{X}, \mathbf{A})$ to $(\mathbf{P}\mathbf{X}, \mathbf{P}\mathbf{A}\mathbf{P}^T)$. The geometric deep-learning prescription is that a node-wise GNN F be permutation equivariant, whereas a

graph-wise scalar readout f be permutation invariant:

$$F(\mathbf{P}\mathbf{X}, \mathbf{P}\mathbf{A}\mathbf{P}^T) = \mathbf{P}F(\mathbf{X}, \mathbf{A}), \quad (9)$$

$$f(\mathbf{P}\mathbf{X}, \mathbf{P}\mathbf{A}\mathbf{P}^T) = f(\mathbf{X}, \mathbf{A}). \quad (10)$$

For an interatomic potential, the first relation applies to node representations and the second to the total energy. Periodic-image and edge features must be related consistently with their incident nodes.

2.1.3 Message passing and many-body representations

In the notation of Ref. [1], a message-passing GNN computes a latent node feature $\mathbf{h}_u$ by applying a learnable message function ψ on every incoming edge, combining the messages with a permutation-invariant aggregation $\bigoplus$, and applying a learnable update function ϕ :

$$\mathbf{h}_u = \phi\left(\mathbf{x}_u, \bigoplus_{(v, \mathbf{n}) \in \mathcal{N}_u} \psi(\mathbf{x}_u, \mathbf{x}_v, \mathbf{x}_{uv}^{\text{edge}})\right). \quad (11)$$

Equation (11) extends the book’s node-feature equation by conditioning ψ on the geometric edge feature, as required for an attributed atomistic graph. A sum is the usual choice for $\bigoplus$, though means, maxima, and other symmetric multiset functions are possible. For a deep GNN, take $\mathbf{h}_u^{(0)} = \mathbf{x}_u^{(0)}$, apply Eq. (11) recursively with layer-specific ψ_ℓ and ϕ_ℓ , and denote each layer’s output by $\mathbf{h}_u^{(\ell+1)}$. Although the graph initially contains only pairwise edges, repeated updates create many-body features: after L layers, an atom can depend on paths extending up to L graph hops. A shared readout ρ maps the final node features to local energies, $\varepsilon_u = \rho(\mathbf{h}_u^{(L)})$, whose sum gives the invariant extensive energy in Eq. (2).

2.1.4 Point Edge Transformer architecture

The Point Edge Transformer (PET) introduced by Pozdnyakov and Ceriotti [5] is the architecture underlying the later PET-MAD potential [4]. PET preserves a distinct latent message on each directed edge. For a central atom i , its transformer processes the incoming edge messages together and produces a separate outgoing message for each neighbor. This avoids compressing the entire neighborhood into a single atom representation before forming outgoing messages. Attention still performs weighted sums internally; the distinction is that the edge identities survive the update.

In the original formulation (Ref. [5], Appendix A), neighbor species, relative Cartesian position, and an incoming message $\mathbf{m}_{j \rightarrow i}^{(\ell-1)} \in \mathbb{R}^{d_{\text{PET}}}$ are encoded as

$$\mathbf{p}_{ij}^{(\ell)} = \text{SiLU}(\mathbf{W}_r^{(\ell)} \mathbf{r}_{ij} + \mathbf{b}_r^{(\ell)}), \quad (12)$$

$$\mathbf{t}_{ij}^{(\ell)} = \text{MLP}_\ell(\mathbf{m}_{j \rightarrow i}^{(\ell-1)} \parallel \mathbf{E}_n^{(\ell)}[Z_j] \parallel \mathbf{p}_{ij}^{(\ell)}), \quad (13)$$

where $\parallel$ denotes concatenation. Each ingredient has width d_{PET} , and the MLP compresses their concatenation from $3d_{\text{PET}}$ to d_{PET} . In the first block the

incoming message is absent, giving $2d_{\text{PET}}$ input features. A separate central token, $\mathbf{t}_{i0}^{(\ell)} = \mathbf{E}_c^{(\ell)}[Z_i]$, uses a distinct species embedding. These are latent feature vectors with no prescribed rotation law. Later PET implementations can modify the encoding and feature dimensions; the project-specific settings are given in Section 4.2.

A transformer mixes the central and neighbor tokens within one environment. For one attention head, with a, b indexing these tokens,

$$\mathbf{q}_a = \mathbf{W}_Q \mathbf{t}_a, \quad \mathbf{k}_b = \mathbf{W}_K \mathbf{t}_b, \quad (14)$$

$$\mathbf{v}_b = \mathbf{W}_V \mathbf{t}_b, \quad (15)$$

$$\alpha_{ab} = \frac{\exp(\mathbf{q}_a^T \mathbf{k}_b / \sqrt{d_k}) c_b}{\sum_c \exp(\mathbf{q}_a^T \mathbf{k}_c / \sqrt{d_k}) c_c}, \quad (16)$$

$$\mathbf{y}_a = \sum_b \alpha_{ab} \mathbf{v}_b. \quad (17)$$

where $c_0 = 1$ for the central token and $c_j = f_c(r_{ij})$ for neighbor j . The cutoff enters before normalization, so a departing neighbor vanishes smoothly from both the weighted sum and its denominator. Multi-head attention repeats this operation in learned subspaces, followed by feed-forward layers and residual connections. No arbitrary neighbor order enters the calculation, so permuting neighbors permutes their output tokens consistently.

Attention makes each neighbor token depend on the other neighbors, allowing many-body interactions within a single message-passing block. Adding attention layers increases local expressive power without extending that block’s neighborhood. Adding message-passing blocks communicates information between environments and enlarges the receptive field. These are separate architecture choices: the water benchmark in Ref. [5] shows that deeper local transformers can outperform a larger number of shallow message-passing blocks at fixed total attention-layer count. Local attention is quadratic in the number of tokens per environment, while the total cost remains approximately linear in atom count for a fixed architecture and bounded neighborhood size.

The output neighbor tokens supply outgoing messages, updated with residual connections to preceding messages. Each block also contributes to the energy. For the summed-edge variant in the original paper,

$$U_\theta = \sum_i b_{Z_i} + \sum_{\ell=1}^L \sum_i \left[H_c^{(\ell)}(\mathbf{x}_{i0}^{(\ell)}) + \sum_{j \in \mathcal{N}_i} f_c(r_{ij}) H_n^{(\ell)}(\mathbf{x}_{ij}^{(\ell)}) \right], \quad (18)$$

where $\mathbf{x}_{i0}$ and $\mathbf{x}_{ij}$ are central and neighbor output tokens, H_c and H_n are learned readout heads, and b_Z are fitted species baselines. Appendix A of Ref. [5] also describes a cutoff-weighted average of edge predictions. The edge terms are many-body learned contributions, not physical pair energies. Contributions from all blocks enter the total, and cutoff weighting is required in the readout as well as in attention.

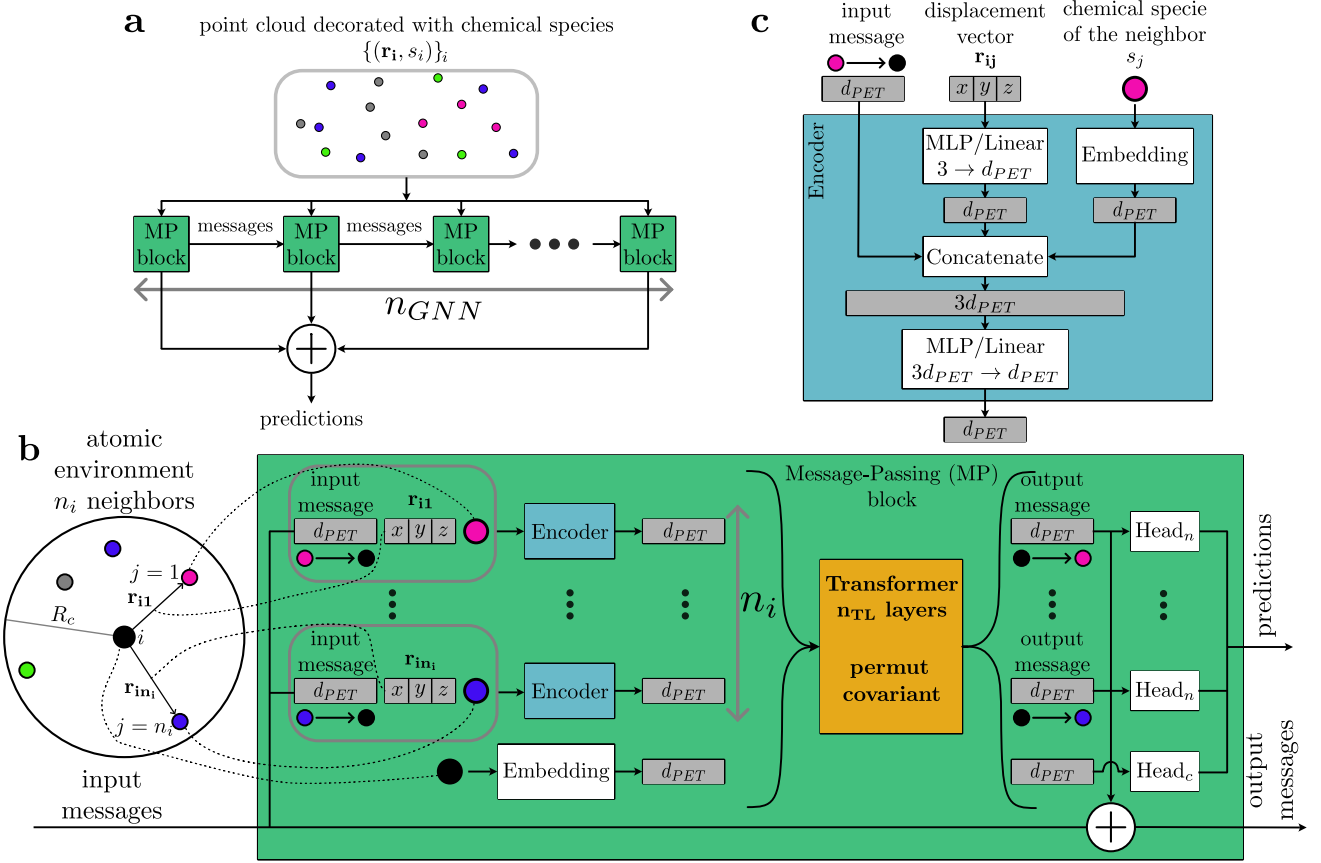


Figure 3: Point Edge Transformer (PET) message-passing architecture. Edge messages associated with a central atom are jointly processed by attention while retaining their individual edge identities for the outgoing update. Figure from the PET paper [5].

2.1.5 Physical symmetries

For an isolated non-magnetic system without external fields, a valid scalar energy should be unchanged by translation, rotation or reflection of the whole structure and by permutation of identical atoms:

$$\begin{aligned} U(\mathbf{R}) &= U(\mathbf{R} + \mathbf{a}) = U(\mathbf{Q}\mathbf{R}) \\ &= U(\mathbf{P}\mathbf{R}), \quad \mathbf{Q} \in O(3). \end{aligned} \quad (19)$$

An invariant scalar is unchanged by a symmetry operation, whereas an equivariant vector changes in the corresponding way; for example, $\mathbf{F}_i(\mathbf{Q}\mathbf{R}) = \mathbf{Q}\mathbf{F}_i(\mathbf{R})$. Translation invariance follows from using relative positions, and the shared local functions and symmetric aggregation enforce permutation invariance. Many MLIPs impose rotational symmetry exactly through invariant geometric descriptors or $E(3)$ -equivariant features. PET instead processes Cartesian displacement vectors with an unconstrained transformer and learns rotational behavior from random rotational augmentation [5, 4]. Thus the PET-MAD backbone is not intrinsically exactly rotation invariant. The PET-MAD publication reports rotational discrepancies usually smaller than its prediction errors [4]; this is not a validation of the particular MOF-5 structures or checkpoint used here. For periodic structures, rotation tests must rotate both positions and cell vectors.

2.1.6 Exact rotational symmetrization with ECSE

The original PET paper also introduces the Equivariant Coordinate System Ensemble (ECSE), a separate construction that imposes exact rotational symmetry on a smooth, translation- and permutation-invariant backbone [5]. Each ordered noncollinear neighbor pair (j, k) defines a local frame $\mathbf{Q}_{ijk}$ that rotates with the atomic environment. For a scalar local prediction, the symmetrized model is

$$\varepsilon_i^{\text{ECSE}} = \frac{\sum_{j \neq k} w_{ijk} \varepsilon_i^0(\mathbf{Q}_{ijk}^T \mathcal{N}_i)}{\sum_{j \neq k} w_{ijk}}, \quad (20)$$

where the notation indicates expressing all relevant displacement vectors in that frame. Vector or tensor predictions are transformed back to the original frame before averaging. The frames in the paper use proper rotations, $SO(3)$; reflection invariance in Eq. (19) is an additional requirement, not a consequence of this construction alone.

Radial weights suppress frames whose defining neighbors approach the cutoff, and angular weights suppress nearly collinear pairs. An adaptive inner radius and smooth pruning reduce the number of frames. Fully collinear environments require a fallback; the paper demonstrates an auxiliary invariant model. Using a smooth ensemble avoids the discontinuities caused by selecting only the two nearest neighbors. This frame-

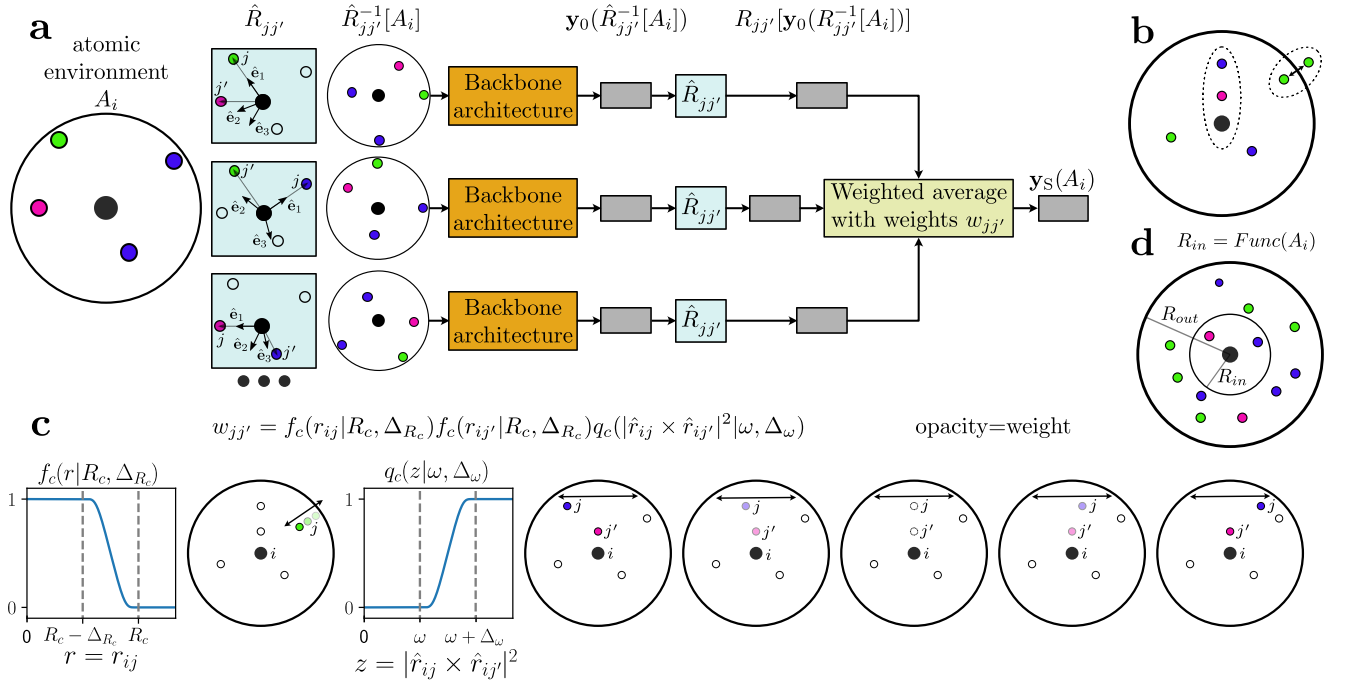


Figure 4: Equivariant Coordinate System Ensemble (ECSE). Local coordinate frames constructed from non-collinear neighbor pairs rotate with the atomic environment; predictions evaluated in these frames are transformed as needed and smoothly averaged. Figure from the PET paper [5].

selection radius is distinct from the adaptive neighbor cutoff in the later PET implementation used here.

Forces and Hessians must differentiate the full expression in Eq. (20), including the position dependence of frames and weights. They cannot be obtained simply by averaging derivatives of the backbone. The paper shows that tight frame-selection parameters can amplify derivative errors even when the energy remains mathematically smooth. Its proof-of-principle ECSE implementation incurs roughly three orders of magnitude of inference overhead (Ref. [5], Section 6 and Appendix F). ECSE is not enabled in this project’s workflow; its exact symmetry properties must therefore be distinguished from those of the deployed PET backbone.

2.1.7 Training and domain of validity

Given reference configurations s , a typical energy–force training objective is

$$\mathcal{L}(\theta) = \sum_s \left[w_E \left(\frac{U_\theta^{(s)} - U_{\text{ref}}^{(s)}}{N_s} \right)^2 + \frac{w_F}{3N_s} \sum_{i\alpha} \left(F_{i\alpha,\theta}^{(s)} - F_{i\alpha,\text{ref}}^{(s)} \right)^2 \right], \quad (21)$$

possibly augmented by stress terms and regularization. Because the force labels are derivatives of the PES, each configuration supplies $3N_s$ local slope constraints in addition to its energy. Optimization adjusts the embeddings, message-passing transformations and readout weights to minimize this loss. The original PET experiments first subtract fitted species baselines and use a loss that rescales energy and force er-

rors by moving validation-set mean squared errors, together with Adam and random rotational augmentation (Ref. [5], Appendix B). Equation (21) is a generic objective, not that paper’s exact normalization. PET-MAD was trained on a chemically diverse DFT dataset rather than on MOF-5 alone, making it a universal pre-trained potential [4].

An MLIP interpolates the reference PES represented by its training data; it does not replace the approximations of the underlying electronic-structure method and need not extrapolate reliably to unfamiliar bonding, charge states, very short distances or thermodynamic phases. Energy and force test errors, trajectory stability and uncertainty indicators are therefore necessary but distinct checks. This project also uses second derivatives, which can be more sensitive than energies or forces and were not directly constrained by the usual loss in Eq. (21). Curvatures and vibrational spectra must therefore be validated explicitly. The original PET benchmarks include isolated distorted methane, water, molecular collisions, crystals, and high-entropy alloys, but do not validate methane adsorption in MOF-5. The paper also reports weaker high-temperature alloy extrapolation despite strong hold-out accuracy, illustrating the distinction between expressiveness and transferability [5].

2.2 Classical molecular dynamics

2.2.1 Hamiltonian dynamics and statistical averages

Molecular dynamics (MD) generates a trajectory through phase space by propagating the nuclear positions and momenta on a chosen PES. For classical

point nuclei with masses m_i , the Hamiltonian is

$$\mathcal{H}(\mathbf{R}, \mathbf{P}) = \sum_{i=1}^N \frac{\mathbf{P}_i^2}{2m_i} + U(\mathbf{R}), \quad (22)$$

and Hamilton's equations give

$$\dot{\mathbf{R}}_i = \frac{\mathbf{P}_i}{m_i}, \quad \dot{\mathbf{P}}_i = -\frac{\partial U}{\partial \mathbf{R}_i} = \mathbf{F}_i. \quad (23)$$

In exact microcanonical dynamics, $\mathcal{H}$ is conserved. If the dynamics is ergodic and the trajectory is sufficiently long, an equilibrium average can be estimated by a time average,

$$\langle A \rangle = \lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^\tau A(\mathbf{R}(t), \mathbf{P}(t)) dt. \quad (24)$$

Finite trajectories contain correlated samples, so their statistical precision is controlled by the simulation length relative to the integrated autocorrelation time, not simply by the number of saved frames.

The continuous equations are replaced by a finite-timestep integrator. A representative symplectic scheme is velocity Verlet,

$$\mathbf{P}_i^{n+1/2} = \mathbf{P}_i^n + \frac{\Delta t}{2} \mathbf{F}_i(\mathbf{R}^n), \quad (25)$$

$$\mathbf{R}_i^{n+1} = \mathbf{R}_i^n + \Delta t \frac{\mathbf{P}_i^{n+1/2}}{m_i}, \quad (26)$$

$$\mathbf{P}_i^{n+1} = \mathbf{P}_i^{n+1/2} + \frac{\Delta t}{2} \mathbf{F}_i(\mathbf{R}^{n+1}). \quad (27)$$

It is time reversible and symplectic, and has global error of order $\mathcal{O}(\Delta t^2)$. It approximately conserves a nearby “shadow” Hamiltonian rather than the exact one. The timestep must resolve the fastest vibration; an apparently controlled temperature does not by itself rule out integration error.

2.2.2 Temperature, ensembles and thermostats

For g unconstrained kinetic degrees of freedom, the instantaneous kinetic temperature is conventionally defined through equipartition,

$$T_{\text{kin}} = \frac{2K}{gk_B}, \quad K = \sum_i \frac{\mathbf{P}_i^2}{2m_i}. \quad (28)$$

Temperature is an ensemble property; T_{kin} fluctuates along a finite canonical trajectory. Constant-energy (NVE) dynamics samples the microcanonical ensemble. To sample the canonical ensemble, whose stationary phase-space density is

$$\rho_{NVT}(\mathbf{R}, \mathbf{P}) \propto \exp[-\beta \mathcal{H}(\mathbf{R}, \mathbf{P})], \quad \beta = (k_B T)^{-1}, \quad (29)$$

the physical variables may be coupled to a thermostat.

For example, Langevin dynamics adds friction and random forces,

$$m_i \ddot{\mathbf{R}}_i = \mathbf{F}_i - m_i \gamma \dot{\mathbf{R}}_i + \boldsymbol{\eta}_i(t), \quad (30)$$

$$\langle \eta_{i\alpha}(t) \eta_{j\beta}(t') \rangle = 2m_i \gamma k_B T \delta_{ij} \delta_{\alpha\beta} \delta(t - t'). \quad (31)$$

The second relation is the fluctuation–dissipation condition that makes the canonical distribution stationary. Weak friction perturbs the natural dynamics less but exchanges heat slowly; strong friction controls temperature rapidly but can overdamp motion. Nosé–Hoover-chain thermostats instead add a sequence of deterministic extended variables and are designed to generate the same canonical distribution while improving ergodicity over a single Nosé–Hoover variable [8]. Thermostat choice affects sampling efficiency and dynamical correlation functions even when equilibrium averages are unchanged.

An isothermal–isobaric simulation additionally evolves the volume or cell so that the target density is proportional to $\exp[-\beta(\mathcal{H} + PV)]$. Flexible-cell algorithms must include the appropriate cell kinetic variables, Jacobian and virial terms; the Martyna–Tobias–Klein scheme is one such extended-system formulation [7]. Reliable NPT dynamics therefore requires a potential with validated cell derivatives or stresses. NVE, NVT and NPT trajectories answer different thermodynamic questions and should not be interchanged solely because their mean temperatures are similar.

2.3 Thermodynamic definitions

2.3.1 Constant-volume and constant-pressure heat capacities

For a system at equilibrium, the constant-volume and constant-pressure heat capacities are

$$C_V = \left(\frac{\partial \langle E \rangle}{\partial T} \right)_{V,N}, \quad (32)$$

$$C_P = \left(\frac{\partial \langle H \rangle}{\partial T} \right)_{P,N}, \quad H = E + PV, \quad (33)$$

where E is the total internal energy and H is the enthalpy. For a stable single phase,

$$C_P - C_V = \frac{TV\alpha_V^2}{\kappa_T}, \quad (34)$$

where $\alpha_V = V^{-1}(\partial V / \partial T)_P$ is the volumetric thermal expansion coefficient and $\kappa_T = -V^{-1}(\partial V / \partial P)_T$ is the isothermal compressibility. Equation (34) explains why C_V and C_P need not be interchangeable even when their numerical difference is small for a particular solid.

In an equilibrated classical canonical ensemble, C_V can also be obtained from total-energy fluctuations,

$$C_V = \frac{\langle E^2 \rangle - \langle E \rangle^2}{k_B T^2}. \quad (35)$$

The analogous isothermal–isobaric expression uses enthalpy fluctuations. These relations require a stationary ensemble, long sampling, and uncertainty estimation that accounts for time correlation. Moreover, classical nuclear dynamics gives equipartition rather than the quantum freezing of high-frequency modes. The present workflow obtains its classical constant-pressure term by differentiating replica-averaged NPT

enthalpies on a temperature grid. It obtains the quantum correction separately from harmonic normal-mode frequencies.

2.3.2 LLPR model uncertainty and CEA propagation

The uncertainty of a machine-learned potential is different from the sampling error of a finite MD trajectory. This project estimates the former for the classical NPT term with a last-layer prediction-rigidity (LLPR) ensemble [9], and the latter with autocorrelation-aware statistics and independent replicas. These quantities are kept separate throughout the analysis.

For a fixed trained representation, write the central PET energy as $\bar{V}(\mathbf{x}) = \mathbf{w}_0^T \mathbf{f}(\mathbf{x})$, where $\mathbf{f}$ is the last-layer feature vector. If $\mathbf{F}$ contains the corresponding features of a reference-labeled covariance set, the meta-train LLPR standard deviation has the form

$$\sigma_{\text{LLPR}}^2(\mathbf{x}) = \alpha^2 \mathbf{f}(\mathbf{x})^T \times (\mathbf{F}^T \mathbf{F} + \lambda \mathbf{I})^{-1} \mathbf{f}(\mathbf{x}). \quad (36)$$

The regularizer λ stabilizes weakly constrained feature directions. The scale α is fitted against energy residuals on a separate, reference-labeled calibration set. The two sets must be disjoint and use the same electronic-structure definition as the base potential. Model predictions cannot serve as reference labels for calibrating their own errors.

Equation (36) is exported as `energy_uncertainty`, a standard deviation in energy units. That scalar is useful for calibration and out-of-domain diagnostics but cannot be propagated through an equilibrium average by itself. The propagation needs signed, correlated changes in energy across configurations. Metatraining therefore samples persistent shallow readout weights,

$$\mathbf{w}_i \sim \mathcal{N}[\mathbf{w}_0, \alpha^2 (\mathbf{F}^T \mathbf{F} + \lambda \mathbf{I})^{-1}], \quad (37)$$

$$V^{(i)}(\mathbf{x}) = \mathbf{w}_i^T \mathbf{f}(\mathbf{x}),$$

and exports their energies as `energy_ensemble`. In the large-member limit, their framewise standard deviation approaches the analytical LLPR standard deviation. This is a shallow ensemble: the learned representation is fixed, so it cannot reveal a model bias or representation error shared by all members and should not be described as a committee of independently trained neural networks.

The ensemble is propagated using the reweighting framework of Imbalzano *et al.* [10]. On a production frame t sampled by the central model at temperature T , define

$$\Delta V_t^{(i)} = V_t^{(i)} - \bar{V}_t, \quad (38)$$

$$H_t^{(i)} = K_t + V_t^{(i)} + P_{\text{ext}} \mathcal{V}_t, \quad (39)$$

where K_t and cell volume $\mathcal{V}_t$ come from the central NPT trajectory. Normalized exponential reweighting gives

$$\langle H^{(i)} \rangle_i^{\text{direct}} = \frac{\sum_t e^{-\beta \Delta V_t^{(i)}} H_t^{(i)}}{\sum_t e^{-\beta \Delta V_t^{(i)}}}, \quad (40)$$

but can collapse when the energy difference is extensive. The primary large-system estimate is therefore the first-order cumulant expansion approximation (CEA),

$$\begin{aligned} \langle H^{(i)} \rangle_i^{\text{CEA}} &= \langle H^{(i)} \rangle_{\bar{V}} \\ &\quad - \beta \text{cov}_{\bar{V}}(H^{(i)}, \Delta V^{(i)}), \end{aligned} \quad (41)$$

$$\beta = (k_B T)^{-1}.$$

Direct reweighting is retained as a diagnostic. Its member-wise effective sample count $N_{\text{eff}} = 1/\sum_t \tilde{w}_t^2$ detects weight collapse. The additional diagnostic $\text{var}_t(\beta \Delta V_t^{(i)})$ should be much smaller than one for a controlled first-order CEA. A large value warns that omitted higher-order terms may be important; agreement with direct reweighting is meaningful only where N_{eff} remains adequate.

At each temperature, independent replicas are averaged member by member. The same exported ensemble and member index must be used at every temperature. The analysis then differentiates each complete enthalpy curve,

$$C_{P,\text{cl}}^{(i)}(T) = \frac{d\langle H^{(i)} \rangle_i^{\text{CEA}}}{dT},$$

$$\sigma_{\text{LLPR},CP}(T) = \left\{ \frac{1}{M-1} \sum_{i=1}^M \left[C_{P,\text{cl}}^{(i)}(T) - \bar{C}_{P,\text{cl}}(T) \right]^2 \right\}^{1/2}. \quad (42)$$

The reported LLPR error is therefore the sample standard deviation of the final member-resolved C_P curves. It is not the instantaneous energy spread, and it is not divided by $\sqrt{M}$: ensemble members sample a predictive distribution rather than provide repeated measurements of one mean.

2.3.3 Why classical and quantum predictions differ

In the classical harmonic limit, every quadratic vibrational degree of freedom contributes k_B to C_V , producing the Dulong–Petit limit at all temperatures. Quantum mechanically, a mode of angular frequency ω becomes thermally populated only when $\hbar\omega$ is comparable to or smaller than $k_B T$. High-frequency framework and molecular modes are therefore partially frozen near room temperature. Classical MD can still describe anharmonic structural motion, but it cannot generate the correct quantum populations simply by extending the trajectory.

3 Reference evidence and the hybrid approximation

Kapil *et al.* studied empty MOF-5 and cells containing 50, 100, and 150 methane molecules [2]. Their results establish the physical motivation for combining a classical anharmonic term with a harmonic quantum correction, and test that approximation for their force field. This project uses the hybrid method as a computational strategy for PET, rather than as a recipe for reproducing their numerical results.

3.1 Differences from the reference calculation

The reference work used analytical QuickFF force fields derived from electronic-structure calculations, not PET-MAD [6]. Framework covalent terms were fitted to B3LYP cluster calculations, with MBIS charges obtained from a periodic PBE density. Nonbonded interactions combined Gaussian-distributed electrostatics with an MM3 Buckingham model and a 15 Å cut-off with tail corrections. Methane was inserted with RASPA, configurations with unrealistic contacts were rejected, and canonical Monte Carlo was used before dynamics. The present workflow instead uses PET for all energy, force, and Hessian evaluations, and independently generated loaded configurations. These differences are expected and are not defects to be removed by matching individual reference parameters.

The reference heat-capacity curves were evaluated from neighboring enthalpies,

$$C_P(T) \simeq \frac{\mathcal{H}(T + \Delta T) - \mathcal{H}(T - \Delta T)}{2\Delta T}, \quad (43)$$

$\Delta T = 25 \text{ K},$

rather than from a single short trajectory. The present classical term uses the same thermodynamic principle: every temperature point depends on converged, statistically independent simulations at adjacent temperatures.

For the harmonic comparison, optimized structures were used to construct Cartesian Hessians with Yaff and normal modes with TAMkin. Empty MOF-5 was tested with both QuickFF and UFF4MOF. The loaded 100-methane calculation used five different Monte Carlo configurations, each optimized into a local minimum before its harmonic spectrum was evaluated.

3.2 Physical conclusions relevant to this project

The main findings of the reference study are:

- Empty MOF-5 is quantum mechanical but nearly harmonic over the studied range. Its quantum harmonic C_V is close to the reported C_P , with a room-temperature gravimetric value of approximately 0.7 J/g/K.
- Classical MD overestimates the empty-framework heat capacity because it populates all vibrational modes according to equipartition.
- Methane loading introduces strongly anharmonic host-guest motion. The reported heat-capacity curves develop a loading-dependent minimum around 200 K, caused mainly by changes in attractive host-guest interactions as confined methane becomes more mobile.
- A harmonic calculation cannot represent diffusion, cage-to-cage motion, or the temperature-dependent localization of methane. Nuclear quantum effects and anharmonic sampling are both needed for a quantitative loaded-system C_P .

- The approximate combination

$$C \approx C_{\text{qn}}^{\text{har}} - C_{\text{cl}}^{\text{har}} + C_{\text{cl}}^{\text{anh}} \quad (44)$$

qualitatively recovers the loaded minimum and becomes useful above roughly 200 K in their 100-methane test.

The operational form used here is Eq. (56). Its classical MD term, its loaded harmonic frequencies, and its mass normalization are all evaluated for the same PET potential and the same loaded composition. The empty-framework Hessian is a reference calculation only. It cannot replace the loaded correction because it omits methane modes and host-guest coupling.

4 Present computational methodology

4.1 Structure preparation and methane insertion

The repository host structure is the conventional cubic MOF-5 cell with 424 atoms and a cell parameter of approximately 25.866 Å. The loaded campaign is generated by `python -m mof_heat_capacity.protocols.loaded`. For every temperature and replica, methane molecules are centered, uniformly randomly rotated using a quaternion, and placed at uniformly sampled fractional coordinates. Periodic atom-atom distances are checked against all previously placed atoms. A 100-methane structure contains 924 atoms. Independent insertion and velocity seeds prevent the nominal replicas from sharing an identical initial state.

This procedure prevents severe initial overlaps, but it is not an adsorption equilibrium calculation. It deliberately does not reproduce the RASPA insertion and Monte Carlo equilibration used by Kapil *et al.*; its adequacy must be shown by independent insertion seeds and sufficient subsequent equilibration before drawing conclusions about loading-dependent thermodynamics.

4.2 PET model and implementation

Section 2.1.4 describes the original PET architecture. The local PET-MAD and PET-SOL conversion metadata both specify the settings in Table 1. These describe the supplied checkpoints, not every PET model; identical architecture settings do not imply identical weights or training data. In particular, the original paper’s default of three attention layers per block differs from the one layer used here.

The locally inspected PyTorch implementation (`metatrain 2026.3.1`, `pet/modules/transformer.py`) embeds four geometric inputs, $(x_{ij}, y_{ij}, z_{ij}, r_{ij})$, rather than only the three Cartesian components in the original Eq. (13). Initial edge messages are species embeddings. Wider central features are contracted to the token width for attention and expanded afterward. With the dimensions in Table 1, each attention head

Table 1: Architecture settings recorded in the local PET-MAD and PET-SOL PET-JAX metadata. The species count is embedding capacity, not a measure of training coverage.

Setting	Value
Outer cutoff	8.0 Å
Cutoff switching width	0.5 Å
Adaptive neighbor-selection setting	40
Message-passing blocks	3
Attention layers per block	1
Token width, d_{PET}	256
Attention heads	8
Transformer feed-forward width	512
Node feature width, d_{node}	1024
Readout feature width, d_{head}	256
Species-embedding capacity	102

has width $256/8 = 32$; $d_{\text{head}}=256$ denotes a readout width. Padded neighbors are masked, and logarithms of cutoff factors are added to attention logits, implementing the weighting in Eq. (17) up to numerical regularization. In the residual feature path of `pet/model.py`, a reverse-neighbor index routes outgoing tokens to receiving environments; averaging the old message with the reversed new output updates the edge state. These are implementation details of the installed PyTorch package; the checkpoint version and PET-JAX conversion must remain compatible with them.

The model weights remain fixed at inference. The repository delegates the network implementation to the PET software stack: its export helper calls `upet.save_upet` to turn the selected checkpoint into a portable metatomic model, which LAMMPS evaluates during production MD. Harmonic analysis converts the matching checkpoint into PET-JAX parameter and metadata files, then uses SADMOF to differentiate its scalar energy. The setup script pins PET-JAX to commit `ed0add4`; conversion transfers parameters rather than retraining a potential. Species baselines have zero Cartesian derivatives at fixed composition, whereas the learned energy scale must be retained in forces and Hessians.

Using matching checkpoints, exports, and conversions is essential to keep MD and harmonic analysis on a consistent PES. This must be checked by comparing energies and forces, including adaptive-cutoff derivative conventions, rather than inferred from matching architecture settings alone. The sparse Hessian path stops adaptive-cutoff derivatives, as described below. The selected metatomic export must expose stress, and campaign configurations record stress validation explicitly rather than inferring capability from a model filename. PET is the architecture; PET-MAD is a general-purpose fitted potential, whose energy, force, stress, and curvature accuracy for MOF-5 and its methane guests must still be validated [4].

4.3 Classical NPT molecular dynamics

The statistical and dynamical foundations of classical MD are summarized in Section 2.2. Only methane-loaded MOF-5 is propagated. LAMMPS evaluates a metatomic PET model in a flexible triclinic cell at an isotropic target pressure of 1 bar. Temperature and pressure are controlled by a Nosé–Hoover-chain/MTTK NPT fix, with chain length 3, thermostat damping time 100 fs, and barostat damping time 1000 fs. The production template specifies 10^6 steps of 0.5 fs, hence 500 ps, saves data every 100 steps (0.05 ps), and discards the first 100 ps. The initial campaign uses one replica at 100, 200, 300, 400, and 500 K. This coarse 100 K spacing reduces cost but limits resolution of curvature in the enthalpy curve and of the heat-capacity minimum; denser temperatures and independent replicas remain convergence extensions.

The exported model must expose stress and must be validated for the intended MOF/loading states before NPT use. Mean loaded-system enthalpies are computed from the stationary portions of the trajectories,

$$\langle H \rangle_T = \langle E + P_{\text{ext}} V \rangle_T, \quad (45)$$

then averaged across replicas and differentiated with respect to temperature. This classical contribution retains guest diffusion, localization changes, and other anharmonic effects sampled by the MLIP.

4.4 Harmonic approximation and Cartesian Hessian

Let $\mathbf{R}_0$ be a stationary reference structure and $\mathbf{u} = \mathbf{R} - \mathbf{R}_0$. Expanding the potential energy gives

$$U(\mathbf{R}_0 + \mathbf{u}) = U(\mathbf{R}_0) + \underbrace{\nabla U(\mathbf{R}_0) \cdot \mathbf{u}}_{=0 \text{ at a minimum}} + \frac{1}{2} \mathbf{u}^T \mathbf{H} \mathbf{u} + \mathcal{O}(u^3), \quad (46)$$

with Cartesian Hessian

$$H_{i\alpha,j\beta} = \frac{\partial^2 U}{\partial R_{i\alpha} \partial R_{j\beta}} = - \frac{\partial F_{i\alpha}}{\partial R_{j\beta}}. \quad (47)$$

For N atoms, $\mathbf{H}$ has dimensions $3N \times 3N$; for the 100-methane cell this is 2772×2772 .

PET-JAX differentiates the model computation graph rather than using displaced finite differences. The implementation uses forward-over-reverse Hessian–vector products,

$$\mathbf{H} \mathbf{v} = \left. \frac{d}{d\epsilon} \nabla U(\mathbf{R} + \epsilon \mathbf{v}) \right|_{\epsilon=0}. \quad (48)$$

Because a local message-passing model couples only graph-connected atoms, the Hessian has a block-sparse structure. SADMOF [11] constructs a graph-derived sparsity pattern and ASDEX colors its columns so that several independent columns can be reconstructed from one differentiated evaluation. The current settings are single precision, graph depth `hops=3`, HVP chunk size 1, and rematerialization enabled. The implementation

calls `get_energy_fn(..., no_shadow=True)`, which stops derivatives through the adaptive cutoff and confines the sparse treatment to the selected graph. Thus this is a specified derivative convention; unrestricted AD through adaptive selection includes additional couplings. Its consistency with the MD force convention requires explicit comparison.

These settings are not established convergence values. In particular, `hops` is the depth of the Hessian sparsity pattern, not the number of PET message-passing blocks. The local SADMOF implementation identifies seven hops as the structurally exact pattern for its three-layer PET-MAD-S graph construction [11]. The project’s three-hop setting is therefore a truncation: it must be enlarged until frequencies and C_V stop changing, and checked against a dense or less aggressively truncated reference where feasible. Precision and the adaptive-cutoff derivative convention also need validation, especially for the low-frequency modes.

4.5 Normal modes and quantum harmonic heat capacity

The Hessian is symmetrized and mass weighted to obtain

$$D_{i\alpha,j\beta} = \frac{H_{i\alpha,j\beta}}{\sqrt{m_i m_j}}, \quad D e_k = \omega_k^2 e_k. \quad (49)$$

The code enforces the force-constant acoustic sum rule before diagonalization,

$$\sum_j H_{i\alpha,j\beta} = 0, \quad (50)$$

so that uniform translations have zero restoring force. The upstream SADMOF observable maps negative eigenvalues to zero frequency for compatibility with its reference heat-capacity convention. The present workflow instead diagonalizes the mass-weighted Hessian locally and retains unstable eigenvalues as signed negative frequencies. Hybrid assembly rejects unexpected imaginary modes and permits at most the three translational near-zero modes, so a thermally displaced frame or saddle point cannot silently enter the correction.

For each retained mode, define

$$x_k = \frac{\hbar \omega_k}{k_B T} = \frac{\hbar c \tilde{\nu}_k}{k_B T}, \quad (51)$$

where $\tilde{\nu}_k$ is the wavenumber. The quantum harmonic contribution is

$$C_{V,k} = k_B \left[\frac{x_k/2}{\sinh(x_k/2)} \right]^2, \quad (52)$$

and the gravimetric result is

$$c_V(T) = \frac{N_A}{M_{\text{cell}}} \sum_{k \in \mathcal{V}} C_{V,k}, \quad (53)$$

in J/g/K. The mass M_{cell} includes methane for a loaded system, so both the spectrum and the normalization change with loading.

One Hessian produces the entire configured 100–500 K curve in 10 K increments. The temperature appearing in Eq. (53) controls oscillator populations; it does not require a separate MD trajectory at every analysis temperature when the force constants are assumed temperature independent.

4.6 Optimized minima and the hybrid correction

The empty-framework calculation starts from the equilibrated MOF-5 structure already available to the project. Its coordinates are relaxed at fixed cell with the selected MLIP and one AD Hessian is evaluated. No empty-MOF MD is needed merely to obtain a Hessian reference.

For the loaded system, final structures from several independent classical MD replicas are quenched at fixed cell with the same MLIP. A Hessian is accepted only after the relaxation reaches the chosen force tolerance and its signed spectrum is checked for unexpected imaginary modes. Raw finite-temperature frames are not treated as normal-mode reference structures. If N_m independent optimized minima are retained, the mean harmonic quantum correction is

$$\overline{\Delta c^{\text{har}}}(T) = \frac{1}{N_m} \sum_{s=1}^{N_m} \left[c_{\text{qn},s}^{\text{har}}(T) - c_{\text{cl},s}^{\text{har}} \right], \quad (54)$$

with sample variance

$$s_{\Delta c}^2(T) = \frac{1}{N_m - 1} \sum_{s=1}^{N_m} \left[\Delta c_s^{\text{har}}(T) - \overline{\Delta c^{\text{har}}}(T) \right]^2. \quad (55)$$

For each minimum, the quantum and classical harmonic terms use exactly the same retained-mode mask. The classical term is k_B per retained mode. The complete loaded-system spectrum is used because framework and methane motion can mix; the empty Hessian alone omits methane quantization and cannot supply this correction.

The final estimator is

$$c_P^{\text{approx}}(T) = c_{P,\text{cl}}^{\text{anh}}(T) + \overline{\Delta c^{\text{har}}}(T), \quad (56)$$

$$c_{P,\text{cl}}^{\text{anh}}(T) = \frac{d\langle H \rangle_{\text{cl}}}{dT}.$$

Extensive terms are combined before division by the total mass of the loaded cell. Replica-aware resampling propagates uncertainty in the enthalpy curve; the spread across optimized minima measures sensitivity of the harmonic correction. The Hessian contribution is formally a C_V correction added to a classical C_P term, following the small solid-state $C_P - C_V$ assumption used in the reference approximation. This mixing and the approximation’s limited validation below roughly 200 K must be stated with the result.

The final standard-uncertainty components are summarized as

$$\sigma_{\text{hyb}}(T) = [\sigma_{\text{MD}}^2(T) + \sigma_{\text{har}}^2(T) + \sigma_{\text{LLPR}, C_P}^2(T)]^{1/2}, \quad (57)$$

where σ_{MD} is the propagated standard error of the central NPT enthalpy curve and σ_{har} is the standard error across independently quenched loaded minima. Equation (57) is a quadrature summary that assumes independence. The three components must also be reported separately. In particular, one trajectory supplies no between-replica uncertainty, and a zero harmonic standard error from one minimum means that contribution was not estimated rather than that it is exactly zero.

5 Results and discussion

The earlier trajectory and raw-frame Hessian calculations were exploratory and are not results of the focused workflow defined here. Their numerical tables and generated overview figure have therefore been removed from this report. Production results should be added only after the new loaded replicas, optimized-minimum Hessians, and hybrid analysis have completed and passed the checks below.

The primary reported result will be a temperature-resolved table or curve containing the replica-averaged classical enthalpy derivative, its MD sampling standard error, its LLPR-CEA model standard deviation, the mean loaded harmonic quantum correction and its minimum-to-minimum standard error, and the combined c_P^{approx} . These error components will be shown separately in addition to any quadrature band. The empty quantum-harmonic curve will be shown as a separately normalized reference. Density, cell response, framework stability, methane mobility, imaginary-mode counts, and relaxation residual forces will support interpretation rather than replace the heat-capacity result. Comparisons with the curves of Kapil *et al.* will be reported as context, not as a target: differences can arise from the potential-energy surface, loading ensemble, and the hybrid approximation itself.

6 Recommended production and analysis protocol

6.1 Empty MOF-5 reference

The equilibrated empty-framework structure is used directly; an empty-MOF MD campaign would add cost without supplying the anharmonic loaded-system term needed by Eq. (56).

1. Relax the atomic coordinates with the same PET-MAD checkpoint used for the Hessian. Report maximum residual force and do not relax the cell unless a stress-capable, validated model is adopted.

2. Check stability using signed eigenvalues and inspect low-frequency modes. A large number of unstable modes invalidates the harmonic interpretation.
3. Converge float32 against float64, graph hops, HVP chunking, acoustic sum-rule treatment, and frequency threshold.
4. Converge the conventional-cell result with respect to supercell size or phonon wave-vector sampling. A single periodic cell Hessian samples only the corresponding finite set of modes.
5. Retain the resulting curve as a separately normalized reference and compare its room-temperature value with the approximately 0.7 J/g/K published reference scale as a contextual check, not a fitting target; PET-MAD replaces QuickFF and its model error remains to be quantified.

6.2 Loaded MOF-5 production and Hessians

For the 100-methane system, the production sequence is:

1. Prepare independent methane placements and velocity seeds for each replica on the enthalpy temperature grid.
2. Run flexible-cell classical NPT only for the loaded system. Determine equilibration from enthalpy, cell, framework, and methane-mobility diagnostics rather than assuming the template duration is sufficient.
3. Average enthalpy over the stationary part of each replica, combine replica uncertainties, and differentiate the temperature dependence.
4. Select representative independent loaded configurations, quench each one to a fixed-cell minimum with the same MLIP, and reject unconverged or unstable minima.
5. Compute one AD Hessian per accepted minimum. Converge precision, graph hops, HVP chunking, acoustic treatment, frequency threshold, and cell size on the quantum-minus-classical correction.
6. Average the loaded corrections, combine them with the classical enthalpy derivative using Eq. (56), and report MD and minimum-to-minimum uncertainties separately as well as jointly.

This calculation applies the hybrid method, rather than attempting an exact reproduction of the reference study. Its success is assessed by internal numerical convergence and by validation of the PET potential for the relevant structures; quantitative accuracy below approximately 200 K remains to be established for the selected MLIP and loading.

6.3 Equilibration and convergence criteria

Production decisions should be based on observables, not a fixed waiting time alone. At minimum, inspect:

- running and block averages of temperature, potential energy, and total energy;
- maximum and distribution of force magnitudes;
- framework bond integrity, pore geometry, and cell consistency;
- methane center-of-mass distributions, host-guest distances, and mean-squared displacements;
- sensitivity to timestep and thermostat friction;
- agreement among independent seeds;
- convergence of the enthalpy derivative with replica count, sampling length, and temperature spacing; and
- convergence of the harmonic correction with independent minima, precision, sparsity depth, cell size, and mode treatment.

The timestep should first be validated in short NVE tests by checking energy drift. Thermostat convergence should then be tested separately, because a thermostat can conceal an integration error while maintaining the desired kinetic temperature.

7 Implementation, outputs, and reproducibility

The production campaign is prepared by `python -m mof_heat_capacity.protocols.loaded` and submitted by `simulation/submit_loaded_md.sh`. These commands create independent loaded structures and LAMMPS/metatomic NPT runs while deliberately rejecting a zero-methane MD campaign. Trajectory diagnostics are submitted with `properties/submit_analysis.sh`.

The command `properties/submit_heat_capacity.sh` takes final structures from selected loaded replicas, relaxes them with `mof_heat_capacity.structures.relax`, and passes the optimized structures to `mof_heat_capacity.analysis.harmonic`. The same command relaxes and evaluates the supplied empty reference directly. Harmonic archives store the run and model identity, optimized structure, temperature grid, frequency array, gravimetric C_V curve, precision, graph hops, HVP chunk size, and rematerialization choice. Finally, `properties/submit_hybrid_analysis.sh` differentiates the loaded NPT enthalpies and writes the classical term, harmonic correction, combined curve, uncertainties, and provenance as NPZ, CSV, and JSON files.

The uncertainty calculations and their stored locations are summarized in Table 2.

The scalar analytical LLPR energy uncertainty is archived but is not used directly as the heat-capacity error. That error is calculated only after CEA has produced a complete enthalpy curve for every persistent ensemble member. The PET-JAX Hessian calculation uses the same persistent LLPR readouts and the hybrid model spread preserves their correlation with the classical CEA term. The Hessians are evaluated at the central-model minima; member-resolved geometry optimization is not included.

Cluster calculations use one node, one task, and one GPU because neither the MD nor Hessian code is MPI or multi-GPU parallel. Independent temperature, replica, or optimized-minimum jobs are the available level of parallelism. Slurm job identifiers and resource use should be retained; elapsed time and peak memory can be recovered with, for example,

```
sacct -j JOB_ID \
--format=JobID,State,Elapsed,\
MaxRSS,AllocTRES
```

Wall time and memory requests should be based on representative jobs. Separate output directories and prefixes are required for every loading, temperature, seed, and replica.

A reproducible result should record:

- source structure, cell, atom count, chemical composition, loading, and methane-insertion seed and rejection criteria;
- PET-MAD checkpoint hash, metatomic export, PET-JAX conversion, software environment, device, and numerical precision;
- LLPR covariance and calibration dataset hashes, energy-label definition, regularizer, calibration method, random seed, member count, exported ensemble hash, frame stride, and CEA diagnostics;
- ensemble, target temperature, timestep, thermostat and friction, number of steps, output stride, velocity seed, equilibration interval, production interval, and restart history;
- Hessian source replica, relaxation state, residual forces, graph hops, HVP chunk size, rematerialization, acoustic sum rule, frequency threshold, and temperature grid; and
- independent-replica count, autocorrelation or block length, uncertainty estimator, and all convergence tests.

8 Limitations and interpretation

The principal limitations of the planned result are:

- PET-MAD and QuickFF describe different potential-energy surfaces. Reproducing the paper’s numerical curves is not a project objective, and any agreement or disagreement requires physical interpretation.

Table 2: Uncertainty components, their calculation locations, and their stored outputs.

Quantity	Where it is calculated	Stored result
LLPR energy standard deviation	metatraining LLPR covariance and calibration	<code>energy_uncertainty</code> ; copied as <code>analytical_energy_uncertainty_eV</code> in the per-run archive
CEA member enthalpies and overlap diagnostics	<code>analysis/uncertainty.py</code> , function	per-run <code>model_uncertainty.npz</code>
Classical LLPR C_P standard deviation	<code>committee_enthalpy_estimates</code> <code>analysis/results.py</code> , after member-wise replica averaging and temperature differentiation	<code>model_uncertainty_heat_capacity.npz</code> and <code>.csv</code>
MD sampling standard error	<code>analysis/statistics.py</code> , then replica combination and derivative resampling in <code>analysis/hybrid.py</code>	<code>classical_anharmonic_cp_standard_error</code>
Harmonic sampling standard error	<code>analysis/hybrid.py</code> , across independent optimized minima	<code>harmonic_quantum_correction_standard_error</code>
Hessian LLPR standard deviation	<code>analysis/harmonic.py</code> , after differentiating persistent LLPR readouts at the central minimum	<code>harmonic_quantum_correction_model_standard_deviation</code>
Combined standard uncertainty	quadrature in <code>analysis/hybrid.py</code>	<code>approximate_cp_combined_standard_uncertainty</code>

- Flexible-cell NPT requires a stress-capable MLIP and a MOF-5-specific stress benchmark; an output capability flag alone is not validation.
- Classical nuclei are used in MD; nuclear quantum effects enter only through the harmonic oscillator formula.
- The random insertion is not an adsorption or Monte Carlo equilibrium distribution.
- A finite number of quenched structures may not represent every thermodynamically important loaded basin. Their spread measures sensitivity but is not automatically a free-energy-weighted average.
- The harmonic correction cannot itself represent methane diffusion or barrier crossing; those effects enter only through classical MD.
- Adding a harmonic C_V correction to a classical C_P term assumes that the corresponding pressure-volume quantum correction is small.
- The approximation was quantitatively successful in the reference 100-methane system mainly above about 200 K; transfer to this MLIP, other loadings, and lower temperatures requires validation.
- Finite-size, sparse-Hessian depth, precision, temperature-derivative, model-validity, and independent-replica errors must all be quantified.
- LLPR varies only the final PET readout and cannot measure bias shared by the frozen representation. CEA is first order, and the present LLPR propagation covers only the classical NPT term, not the harmonic correction.

These qualifications define the boundary between an efficient, physically motivated approximation and

an exact quantum-anharmonic thermodynamic result. They should be carried into every plot, table, and comparison with experiment.

9 Conclusions

The focused workflow avoids unnecessary empty-framework MD. Classical NPT trajectories are reserved for loaded MOF-5, where they are needed to describe anharmonic guest and host-guest motion. Sparse AD Hessians are evaluated only at converged fixed-cell minima and provide the loaded-system quantum-minus-classical harmonic correction. The equilibrated empty structure contributes one direct reference Hessian.

The final quantity is the classical loaded-system enthalpy derivative plus the mean loaded harmonic correction. Its credibility depends on stress validation, stationary and replicated MD, stable optimized minima, numerical Hessian convergence, consistent mode and mass bookkeeping, and explicit uncertainty propagation. This strategy applies the cost-saving hybrid approximation motivated by Kapil *et al.* to the PET potential; it does not seek to reproduce the reference force field, sampling protocol, or numerical curves.

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